

Problem 1. A projectile is launched at an angle of 45° and lands 12 s later at the same height as it was launched.

- (a) What is the initial speed of the projectile?
- (b) What is the maximum altitude?
- (c) What is the range?
- (d) Calculate the displacement from the point of launch to the position on its trajectory at 9 s.

Solution.

Since the projectile lands at the same height from which it was launched, the total time of flight is

$$T = \frac{2v_0 \sin \theta}{g}.$$

Solving for the initial speed gives

$$v_0 = \frac{gT}{2 \sin \theta} = \frac{(9.8 \text{ m/s}^2)(12 \text{ s})}{2 \sin 45^\circ} = 83.2 \text{ m/s}.$$

- (a) The initial speed is

$$\boxed{v_0 = 83.2 \text{ m/s}}.$$

- (b) The maximum altitude is

$$h_{\max} = \frac{(v_0 \sin \theta)^2}{2g}.$$

Since

$$v_0 \sin 45^\circ = 58.8 \text{ m/s},$$

we get

$$h_{\max} = \frac{(58.8 \text{ m/s})^2}{2(9.8 \text{ m/s}^2)} = 176.4 \text{ m}.$$

Thus,

$$\boxed{h_{\max} = 176 \text{ m}}.$$

- (c) The range is

$$R = (v_0 \cos \theta)T.$$

Since

$$v_0 \cos 45^\circ = 58.8 \text{ m/s},$$

we have

$$R = (58.8 \text{ m/s})(12 \text{ s}) = 705.6 \text{ m}.$$

Thus,

$$\boxed{R = 706 \text{ m}}.$$

(d) At $t = 9$ s,

$$x = (v_0 \cos \theta)t = (58.8 \text{ m/s})(9 \text{ s}) = 529.2 \text{ m},$$

and

$$y = (v_0 \sin \theta)t - \frac{1}{2}gt^2 = (58.8)(9) - \frac{1}{2}(9.8)(9)^2 = 132.3 \text{ m}.$$

Therefore, the displacement is

$$\boxed{\Delta \vec{r} = (529 \text{ m})\hat{i} + (132 \text{ m})\hat{j}}.$$

Problem 2. A dense star, with a radius of about 18 km, rotates once every 1.5 s.

- (a) What is the speed of a particle on the star's equator?
- (b) What is the magnitude of the particle's centripetal acceleration?
- (c) If the neutron star rotates faster, do the answers to (a) and (b) increase, decrease, or remain the same?

Solution.

(a)

$$v = \frac{2\pi r}{T} = \frac{2\pi(18 \text{ km})}{1.5 \text{ s}} = 75.4 \text{ km/s} = 7.54 \times 10^4 \text{ m/s}.$$

Thus,

$$\boxed{v = 7.54 \times 10^4 \text{ m/s}}.$$

(b) The magnitude of the acceleration is

$$a = \frac{v^2}{r} = \frac{(75.4 \text{ km/s})^2}{18 \text{ km}} = 3.16 \times 10^5 \text{ m/s}^2.$$

Thus,

$$\boxed{a = 3.16 \times 10^5 \text{ m/s}^2}.$$

(c) Clearly, both v and a will increase if T is reduced.

Problem 3. A 72 kg crate is dragged across a floor by pulling on a rope attached to the crate and inclined 20° above the horizontal.

- (a) If the coefficient of static friction is 0.45, what minimum force magnitude is required from the rope to start the crate moving?
- (b) If $\mu_k = 0.30$, what is the magnitude of the initial acceleration of the crate?

Solution.

The free-body diagram for the crate contains the tension force $\vec{T}$ of the rope on the crate, the normal force $\vec{F}_N$ of the floor on the crate, gravity $m\vec{g}$, and the friction force $\vec{f}$. We take the $+x$ direction to be horizontal to the right and the $+y$ direction to be up. We assume the crate is motionless. The equations for the x and y components of the force according to Newton's second law are:

$$\begin{aligned} T \cos \theta - f &= 0, \\ T \sin \theta + F_N - mg &= 0, \end{aligned}$$

where $\theta = 20^\circ$ is the angle between the rope and the horizontal. The first equation gives $f = T \cos \theta$ and the second gives $F_N = mg - T \sin \theta$. If the crate is to remain at rest, f must be less than $\mu_s F_N$. When the tension force is sufficient to just start the crate moving, we must have

$$T \cos \theta = \mu_s (mg - T \sin \theta).$$

We solve for the tension:

$$T = \frac{\mu_s mg}{\cos \theta + \mu_s \sin \theta} = \frac{(0.45)(72 \text{ kg})(9.8 \text{ m/s}^2)}{\cos 20^\circ + 0.45 \sin 20^\circ} = 290 \text{ N}.$$

Thus,

$$\boxed{T = 2.90 \times 10^2 \text{ N}}.$$

For the moving crate, the second law equations are

$$\begin{aligned} T \cos \theta - f &= ma, \\ F_N + T \sin \theta - mg &= 0. \end{aligned}$$

Now $f = \mu_k F_N$, and the second equation gives $F_N = mg - T \sin \theta$, which yields

$$f = \mu_k (mg - T \sin \theta).$$

This expression is substituted into the first equation:

$$T \cos \theta - \mu_k (mg - T \sin \theta) = ma.$$

Therefore,

$$a = \frac{T(\cos \theta + \mu_k \sin \theta)}{m} - \mu_k g.$$

Numerically,

$$a = \frac{(290 \text{ N})(\cos 20^\circ + 0.30 \sin 20^\circ)}{72 \text{ kg}} - (0.30)(9.8 \text{ m/s}^2) = 1.26 \text{ m/s}^2.$$

Thus,

$$\boxed{a = 1.3 \text{ m/s}^2}.$$

Problem 4. Block 1 of mass $m_1 = 2.5$ kg and block 2 of mass $m_2 = 4.0$ kg are connected by a string of negligible mass and are initially held in place. Block 2 is on a frictionless surface tilted at $\theta = 35^\circ$. The coefficient of kinetic friction between block 1 and the horizontal surface is 0.20. The pulley has negligible mass and friction. Once they are released, the blocks move. What then is the tension in the string?

Solution.

Take the motion of block 2 down the incline as positive, so block 1 moves horizontally toward the pulley. For block 1, the tension pulls forward and kinetic friction opposes the motion:

$$T - f_k = m_1 a.$$

Since block 1 is on a horizontal surface,

$$f_k = \mu_k m_1 g.$$

Thus,

$$T - \mu_k m_1 g = m_1 a.$$

For block 2, the component of gravity down the incline is $m_2 g \sin \theta$, and the tension points up the incline:

$$m_2 g \sin \theta - T = m_2 a.$$

Adding the two equations gives

$$m_2 g \sin \theta - \mu_k m_1 g = (m_1 + m_2) a.$$

Therefore,

$$a = \frac{m_2 g \sin \theta - \mu_k m_1 g}{m_1 + m_2}.$$

Substituting the values:

$$a = \frac{(4.0 \text{ kg})(9.8 \text{ m/s}^2) \sin 35^\circ - (0.20)(2.5 \text{ kg})(9.8 \text{ m/s}^2)}{2.5 \text{ kg} + 4.0 \text{ kg}} = 2.71 \text{ m/s}^2.$$

Now use the equation for block 1:

$$T = m_1 a + \mu_k m_1 g.$$

Thus,

$$T = (2.5 \text{ kg})(2.71 \text{ m/s}^2) + (0.20)(2.5 \text{ kg})(9.8 \text{ m/s}^2) = 11.7 \text{ N}.$$

Therefore,

$$\boxed{T = 12 \text{ N}}.$$

Problem 5. A sled and its rider, with a total mass of 90 kg, emerge from a downhill track onto a horizontal straight track with an initial speed of 32 m/s. If a force slows them to a stop at a constant rate of 1.6 m/s²,

- (a) What magnitude F is required for the force?
- (b) What distance d do they travel while slowing?
- (c) What work W is done on them by the force?

Solution.

We choose $+x$ as the direction of motion, so $\vec{a}$ and $\vec{F}$ are negative-valued.

- (a) Newton's second law gives

$$\vec{F} = m\vec{a} = (90 \text{ kg})(-1.6 \text{ m/s}^2).$$

Therefore,

$$F = |\vec{F}| = 144 \text{ N}.$$

Thus,

$$\boxed{F = 1.44 \times 10^2 \text{ N}}.$$

- (b) From

$$v^2 = v_0^2 + 2a\Delta x,$$

with $v = 0$, we have

$$0 = (32 \text{ m/s})^2 + 2(-1.6 \text{ m/s}^2)\Delta x.$$

Thus,

$$\Delta x = -\frac{(32 \text{ m/s})^2}{2(-1.6 \text{ m/s}^2)} = 320 \text{ m}.$$

Therefore,

$$\boxed{d = 320 \text{ m}}.$$

- (c) Since the force is opposite to the direction of motion,

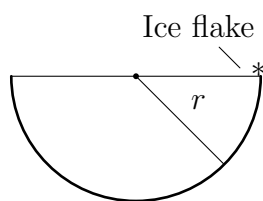
$$W = -Fd.$$

Therefore,

$$W = -(144 \text{ N})(320 \text{ m}) = -4.61 \times 10^4 \text{ J}.$$

Thus,

$$\boxed{W = -4.61 \times 10^4 \text{ J}}.$$

Problem 6.

In the figure on the left, a 3.00 g ice flake is released from the edge of a hemispherical bowl whose radius r is 18.0 cm. The flake–bowl contact is frictionless.

- How much work is done on the flake by the gravitational force during the flake's descent to the bottom of the bowl?
- What is the change in the potential energy of the flake–Earth system during that descent?
- If that potential energy is taken to be zero at the bottom of the bowl, what is its value when the flake is released?
- If, instead, the potential energy is taken to be zero at the release point, what is its value when the flake reaches the bottom of the bowl?
- If the mass of the flake were doubled, would the magnitudes of the answers to (a) through (d) increase, decrease, or remain the same?
- What is the speed of the flake when it reaches the bottom of the bowl?
- If we substituted a second flake with twice the mass, what would its speed be?
- If, instead, we gave the flake an initial downward speed along the bowl, would the answer to (f) increase, decrease, or remain the same?

Solution.

- (a) The force of gravity is constant, so the work it does is given by

$$W = \vec{F} \cdot \vec{d}.$$

The force is vertically downward and has magnitude mg , so this reduces to

$$W = mgh.$$

Here the vertical height is equal to the radius r of the bowl. Thus,

$$W = mgr = (3.00 \times 10^{-3} \text{ kg})(9.8 \text{ m/s}^2)(18.0 \times 10^{-2} \text{ m}) = 5.29 \times 10^{-3} \text{ J}.$$

Therefore,

$$\boxed{W = 5.29 \times 10^{-3} \text{ J}}.$$

- (b) The force of gravity is conservative, so the change in gravitational potential energy is the negative of the work done:

$$\Delta U = -W = -5.29 \times 10^{-3} \text{ J}.$$

Thus,

$$\boxed{\Delta U = -5.29 \times 10^{-3} \text{ J}}.$$

- (c) If $U = 0$ at the bottom, then the potential energy at the release point is

$$U = mgr = 5.29 \times 10^{-3} \text{ J}.$$

Thus,

$$\boxed{U_{\text{top}} = +5.29 \times 10^{-3} \text{ J}}.$$

- (d) If $U = 0$ at the release point, then the potential energy at the bottom is lower by mgr :

$$U_{\text{bottom}} = -5.29 \times 10^{-3} \text{ J}.$$

Thus,

$$\boxed{U_{\text{bottom}} = -5.29 \times 10^{-3} \text{ J}}.$$

- (e) The answers to (a) through (d) are proportional to the mass. If the mass is doubled, the magnitudes of the answers double.
- (f) By conservation of mechanical energy,

$$K_f + U_f = K_i + U_i.$$

Taking the potential energy to be zero at the bottom of the bowl,

$$\frac{1}{2}mv^2 = mgr.$$

The mass cancels, giving

$$v = \sqrt{2gr} = \sqrt{2(9.8 \text{ m/s}^2)(0.180 \text{ m})} = 1.88 \text{ m/s}.$$

Thus,

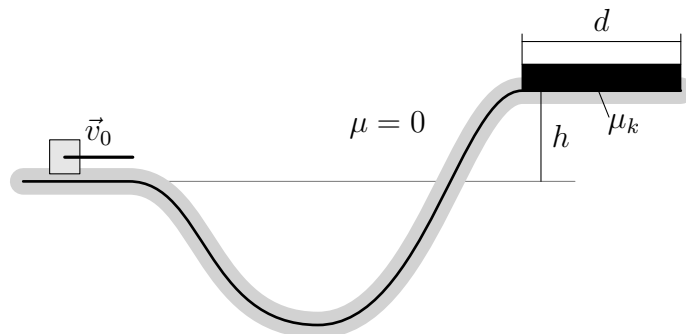
$$\boxed{v = 1.88 \text{ m/s}}.$$

- (g) Since the expression for speed does not contain the mass of the flake, the speed would be the same:

$$\boxed{v = 1.88 \text{ m/s}}.$$

- (h) If the flake is given an initial downward speed, then it starts with nonzero kinetic energy. Therefore, its kinetic energy at the bottom is greater, so the answer to (f) would increase.

Problem 7. A block slides along a track from one level to a higher level after passing through an intermediate valley. The track is frictionless until the block reaches the higher level. There a frictional force stops the block in a distance d . The block's initial speed v_0 is 7.0 m/s, the height difference h is 1.5 m, and μ_k is 0.50. Find d .



Solution.

Since the valley is frictionless, the only reason for the speed being less when it reaches the higher level is the gain in potential energy

$$\Delta U = mgh.$$

Sliding along the rough surface of the higher level, the block finally stops since its remaining kinetic energy has turned to thermal energy:

$$\Delta E_{\text{th}} = f_k d = \mu_k mgd.$$

Thus,

$$K_i = \Delta U + \Delta E_{\text{th}} = mg(h + \mu_k d),$$

where

$$K_i = \frac{1}{2}mv_i^2.$$

Dividing by mass and rearranging, we obtain

$$d = \frac{v_i^2}{2\mu_k g} - \frac{h}{\mu_k}.$$

Substituting the values:

$$d = \frac{(7.0 \text{ m/s})^2}{2(0.50)(9.8 \text{ m/s}^2)} - \frac{1.5 \text{ m}}{0.50}.$$

Thus,

$$d = 5.0 \text{ m} - 3.0 \text{ m} = 2.0 \text{ m}.$$

Therefore,

$$\boxed{d = 2.0 \text{ m}}.$$